

Task families

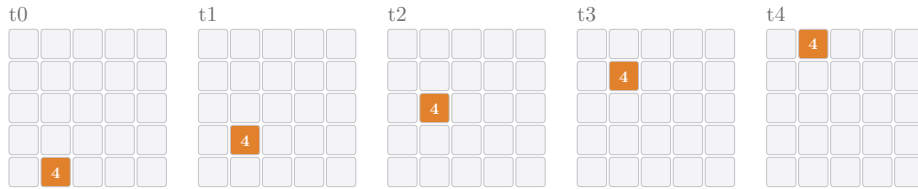
one representative scene each — 5×5 grids

 empty  id 1  id 2  wall  id 4  id 5

Physics — a body moves [physics]

(step v d)

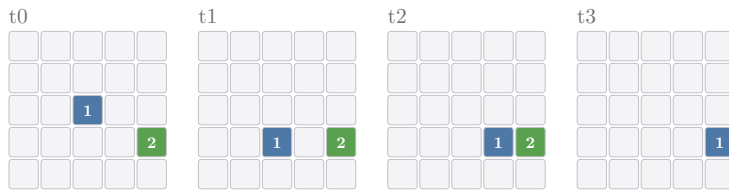
val=4, dir=up



Desire — seek a goal (true belief) [desire]

(optimize (neg_dist gv) av)

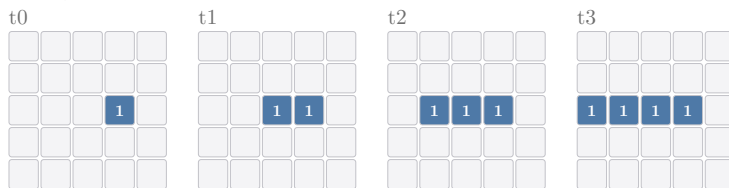
av=1, gv=2



Overlay — motion-blur trail [overlay]

(fork (step v d) overlay)

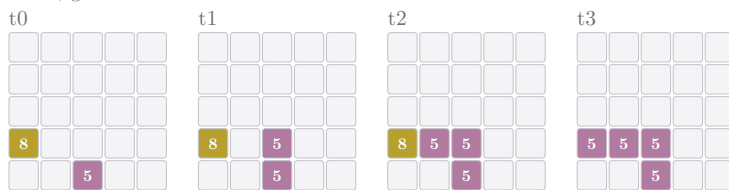
val=1, dir=left



Comet — fork with a seek derive [comet]

(fork (optimize (neg_dist gv) av) overlay)

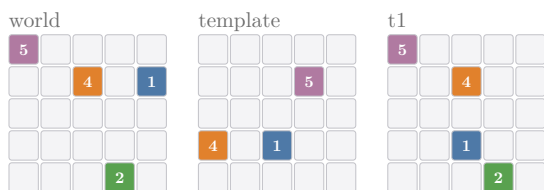
av=5, gv=8



Registration — snap one object [registration]

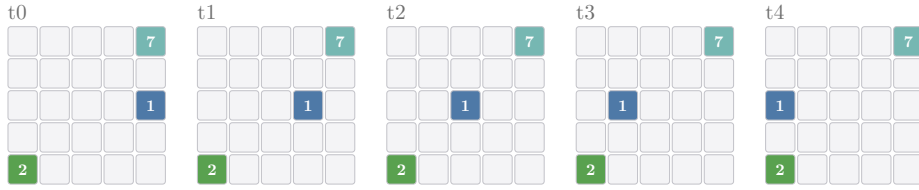
(sync_to_world v)

val=1



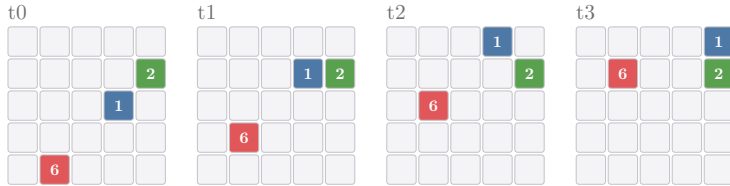
Goal displacement — Sally-Anne [belief_goal]

```
(fork (seq (step gv d) (optimize (neg_dist gv) av)) (sync_to_world av))
av=1, gv=2, displaced_to=(2, 0), dirs=('up', 'up'), distractors={7: (0, 4)}
```



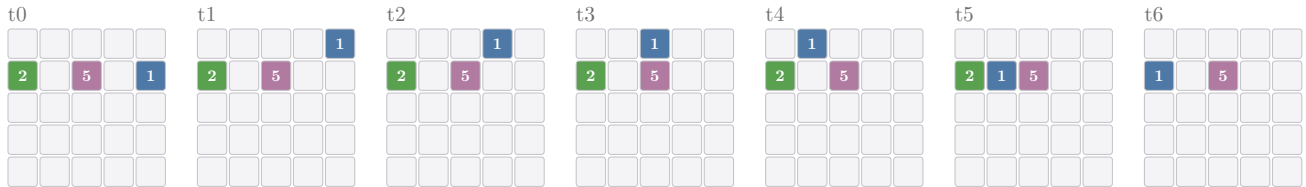
Two observers — one goal, one grid, one fork [belief_observers]

```
(compose (optimize (neg_dist gv) ao) (fork (compose (step gv d) (optimize (neg_dist gv) av))
(sync_to_world av)))
av=1, gv=2, observer=6, displaced_to=(0, 4), dirs=('up',)
```



False belief — detour round an invisible wall [belief_wall]

```
(fork (compose (wall_at r c) (optimize (neg_dist gv) av)) (sync_to_world av))
av=1, gv=2, pw=(1, 2), bv=5
```



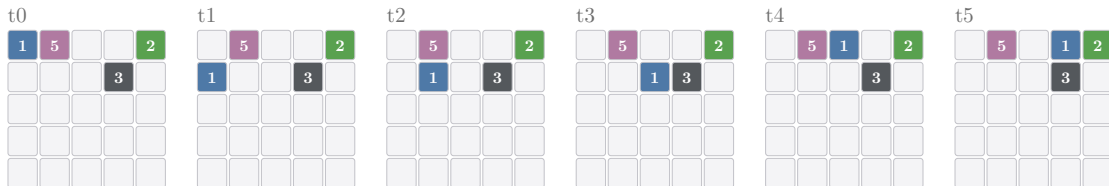
Witness belief — a bystander crosses the phantom wall [belief_witness]

```
(compose (fork (compose (wall_at r c) (optimize (neg_dist gv) av)) (sync_to_world av)) (optimize
(neg_dist gw) aw))
av=1, gv=2, aw=4, gw=5, pw=(1, 3)
```



False obstacle — wrong about wall AND goal [belief_false_obstacle]

```
(fork (seq (wall_at Wb) (step gv d) (optimize (neg_dist gv) av)) (sync_to_world av))
av=1, gv=2, pw=(0, 1), real_wall=(1, 3), bv=5, displaced_to=(0, 3), dir=left
```



Flee — avoid the nearest hazard [flee]

```
(optimize (distance hv) av)
```

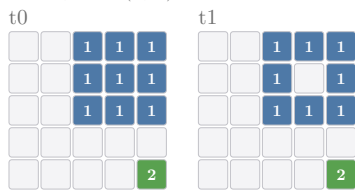
av=1, hv=4



Deletion — punch one hole [deletion]

(clear_at r c)

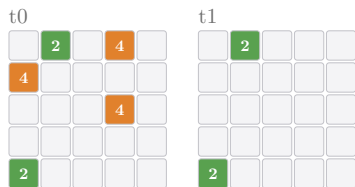
val=1, cell=(1, 3)



Denoise — drop a noise value [denoise]

(erase nv)

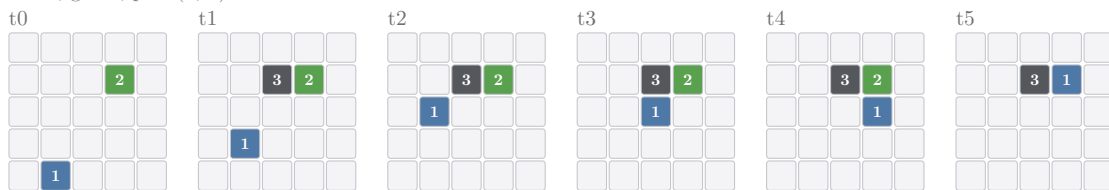
noise=4, signal=2



Obstacle — detour round a REAL wall [obstacle]

(compose (wall_at r c) (optimize (neg_dist gv) av))

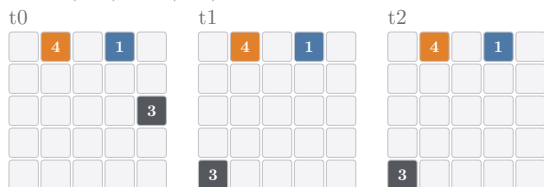
av=1, gv=2, pw=(1, 2)



Relocation — a real wall jumps cell [relocate]

(compose (clear_at r c) (wall_at r' c'))

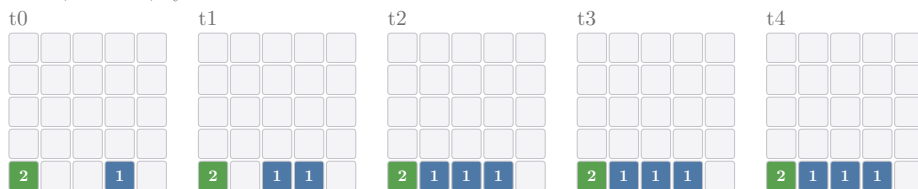
from=(2, 4), to=(4, 0)



Underlay — world-wins motion trail [underlay]

(fork (step v d) underlay)

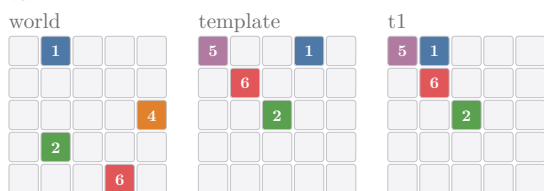
val=1, dir=left, bystander=2



Perception — record into the model [perception]

(sync_to_model v)

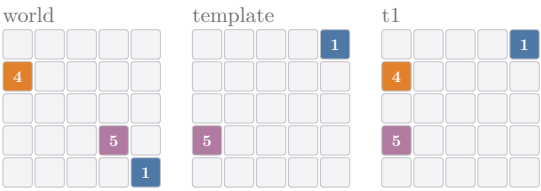
val=1



Multi-registration — snap all [multi_reg]

sync_all

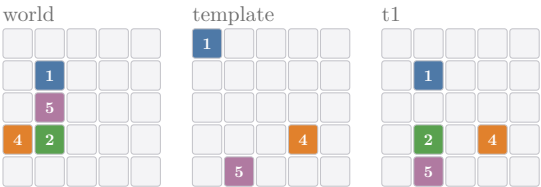
vals=[1, 5]



Registration-except — all but one [reg_except]

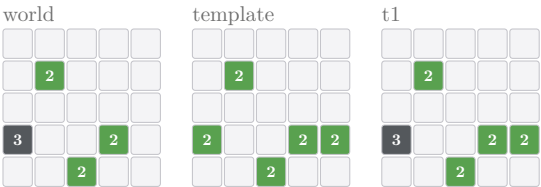
(sync_except a)

anchor=1



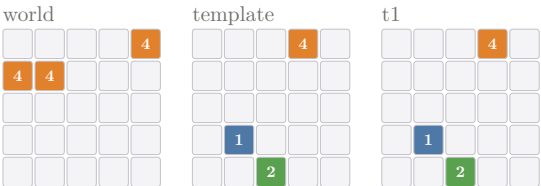
Inpainting — fill holes, keep pixels [inpaint]

underlay



Readout — report the model [readout]

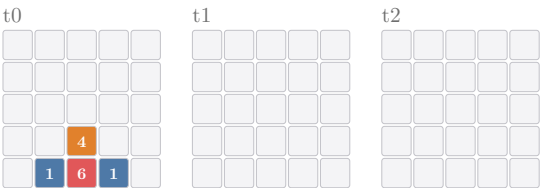
snd_gg



Wipe — the grid goes blank [wipe]

(pipe_gpg pair_blank snd_gg)

vals=[1, 4, 6]



Task families — collapsed path view

the same trajectories on one grid, an arrow per transition

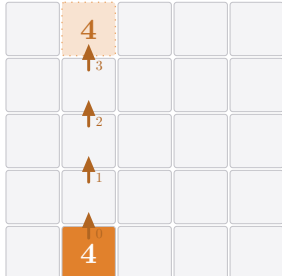
 empty  id 1  id 2  wall  id 4  id 5

Physics — a body moves [physics]

(step v d)

val=4, dir=up

t0→t4 (5 frames)

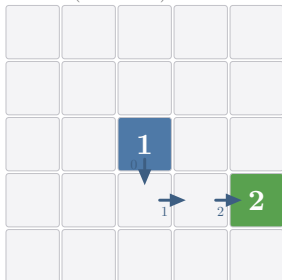


Desire — seek a goal (true belief) [desire]

(optimize (neg_dist gv) av)

av=1, gv=2

t0→t3 (4 frames)

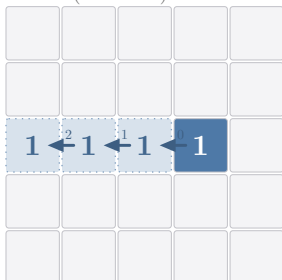


Overlay — motion-blur trail [overlay]

(fork (step v d) overlay)

val=1, dir=left

t0→t3 (4 frames)

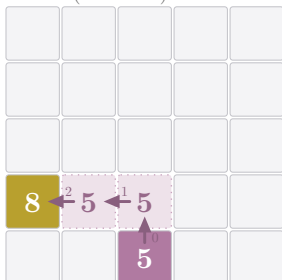


Comet — fork with a seek derive [comet]

(fork (optimize (neg_dist gv) av) overlay)

av=5, gv=8

t0→t3 (4 frames)

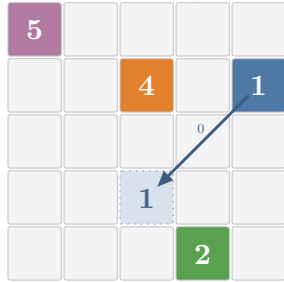


Registration — snap one object [registration]

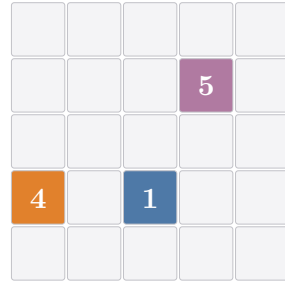
(sync_to_world v)

val=1

world→t1 (2 frames)



template

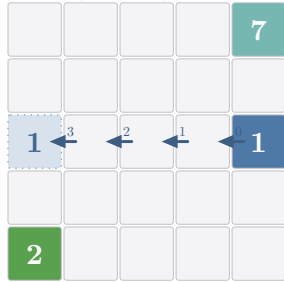


Goal displacement — Sally-Anne [belief_goal]

(fork (seq (step gv d) (optimize (neg_dist gv) av)) (sync_to_world av))

av=1, gv=2, displaced_to=(2, 0), dirs=('up', 'up'), distractors={7: (0, 4)}

t0→t4 (5 frames)

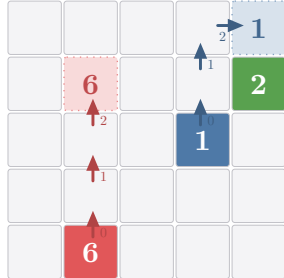


Two observers — one goal, one grid, one fork [belief_observers]

(compose (optimize (neg_dist gv) ao) (fork (compose (step gv d) (optimize (neg_dist gv) av)) (sync_to_world av)))

av=1, gv=2, observer=6, displaced_to=(0, 4), dirs=('up',)

t0→t3 (4 frames)

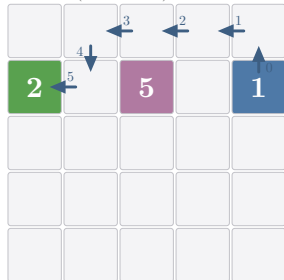


False belief — detour round an invisible wall [belief_wall]

(fork (compose (wall_at r c) (optimize (neg_dist gv) av)) (sync_to_world av))

av=1, gv=2, pw=(1, 2), bv=5

t0→t6 (7 frames)

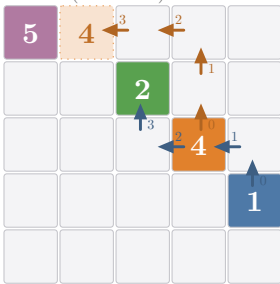


Witness belief — a bystander crosses the phantom wall ^[belief_witness]

(compose (fork (compose (wall_at r c) (optimize (neg_dist gv) av)) (sync_to_world av)) (optimize (neg_dist gw) aw))

av=1, gv=2, aw=4, gw=5, pw=(1, 3)

t0→t4 (5 frames)

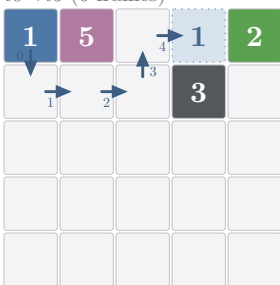


False obstacle — wrong about wall AND goal ^[belief_false_obstacle]

(fork (seq (wall_at Wb) (step gv d) (optimize (neg_dist gv) av)) (sync_to_world av))

av=1, gv=2, pw=(0, 1), real_wall=(1, 3), bv=5, displaced_to=(0, 3), dir=left

t0→t5 (6 frames)

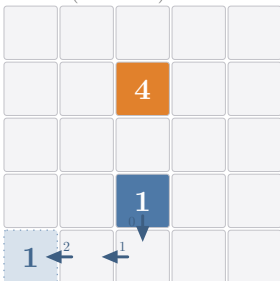


Flee — avoid the nearest hazard ^[flee]

(optimize (distance hv) av)

av=1, hv=4

t0→t4 (5 frames)

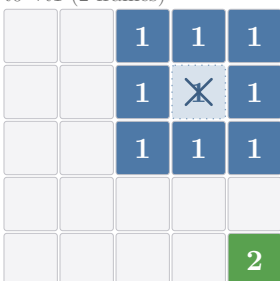


Deletion — punch one hole ^[deletion]

(clear_at r c)

val=1, cell=(1, 3)

t0→t1 (2 frames)

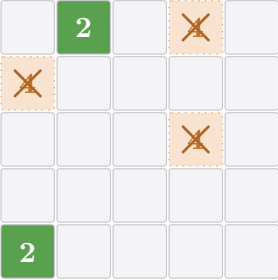


Denoise — drop a noise value [denoise]

(erase nv)

noise=4, signal=2

t0→t1 (2 frames)

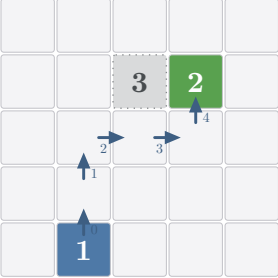


Obstacle — detour round a REAL wall [obstacle]

(compose (wall_at r c) (optimize (neg_dist gv) av))

av=1, gv=2, pw=(1, 2)

t0→t5 (6 frames)

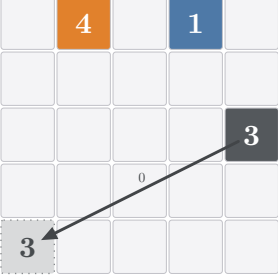


Relocation — a real wall jumps cell [relocate]

(compose (clear_at r c) (wall_at r' c'))

from=(2, 4), to=(4, 0)

t0→t2 (3 frames)



Underlay — world-wins motion trail [underlay]

(fork (step v d) underlay)

val=1, dir=left, bystander=2

t0→t4 (5 frames)

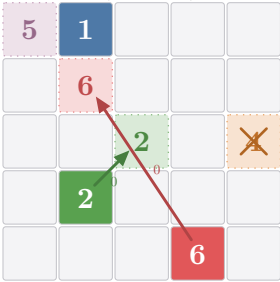


Perception — record into the model [perception]

(sync_to_model v)

val=1

world→t1 (2 frames)



template

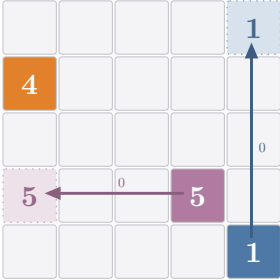


Multi-registration — snap all [multi_reg]

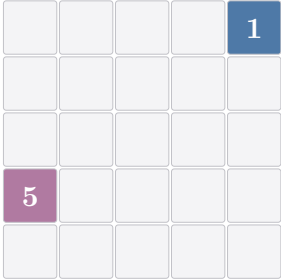
sync_all

vals=[1, 5]

world→t1 (2 frames)



template

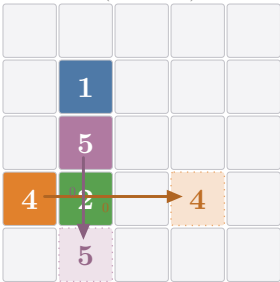


Registration-except — all but one [reg_except]

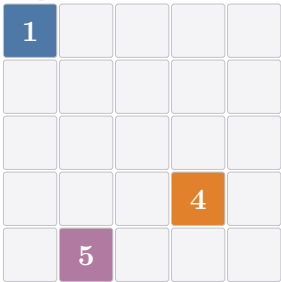
(sync_except a)

anchor=1

world→t1 (2 frames)



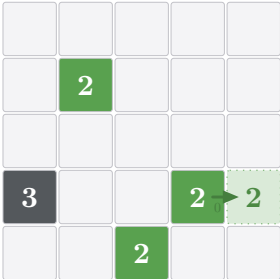
template



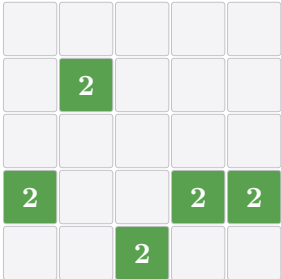
Inpainting — fill holes, keep pixels [inpaint]

underlay

world→t1 (2 frames)



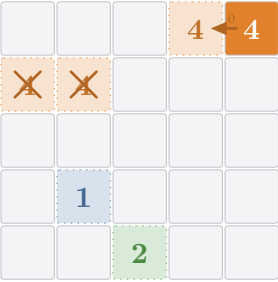
template



Readout — report the model [readout]

snd_gg

world→t1 (2 frames)



template

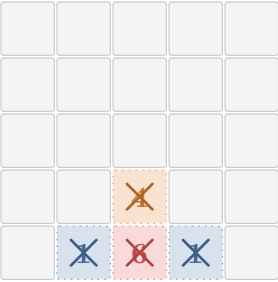


Wipe — the grid goes blank [wipe]

(pipe_gpg pair_blank snd_gg)

vals=[1, 4, 6]

t0→t2 (3 frames)



Task families — the whole scene set

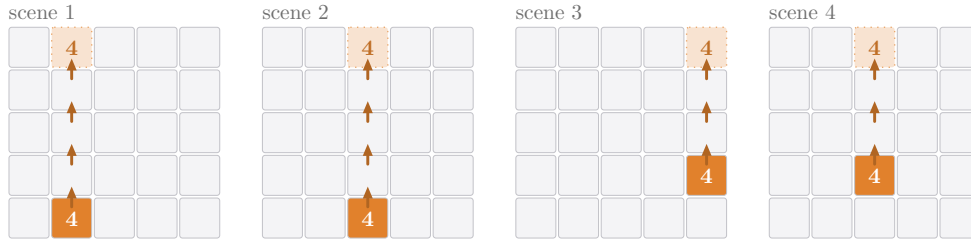
every scene of one task per family, path view: one latent program, k different starts, all of which a solution must reproduce

empty 1 id 1 2 id 2 3 wall 4 id 4 5 id 5

Physics — a body moves [physics]

(step v d)

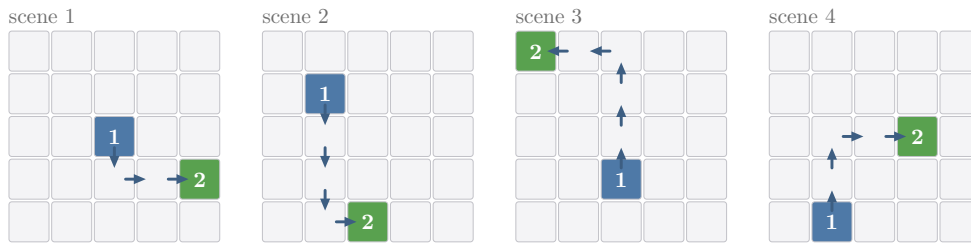
val=4, dir=up



Desire — seek a goal (true belief) [desire]

(optimize (neg_dist gv) av)

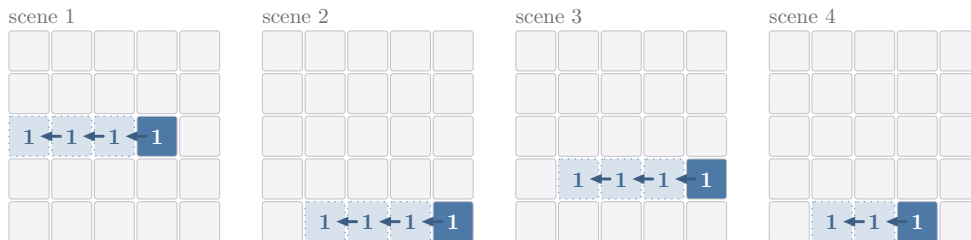
av=1, gv=2



Overlay — motion-blur trail [overlay]

(fork (step v d) overlay)

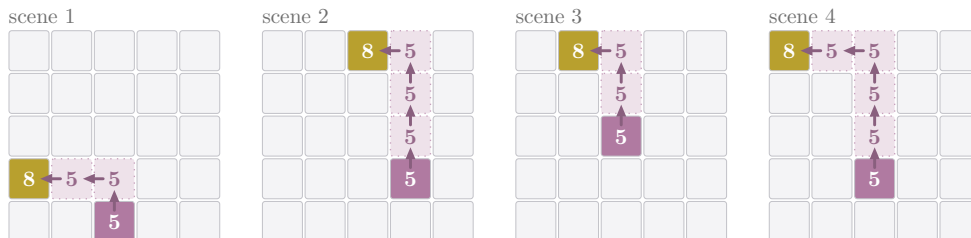
val=1, dir=left



Comet — fork with a seek derive [comet]

(fork (optimize (neg_dist gv) av) overlay)

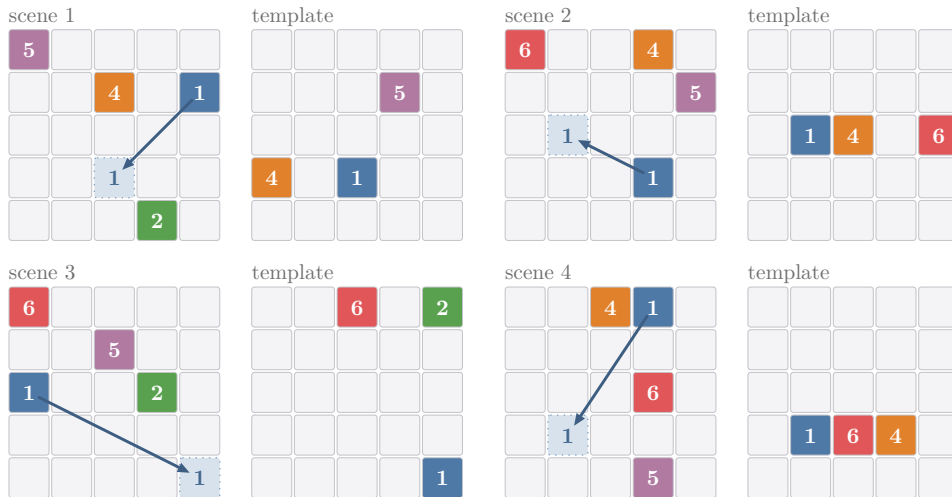
av=5, gv=8



Registration — snap one object [registration]

(sync_to_world v)

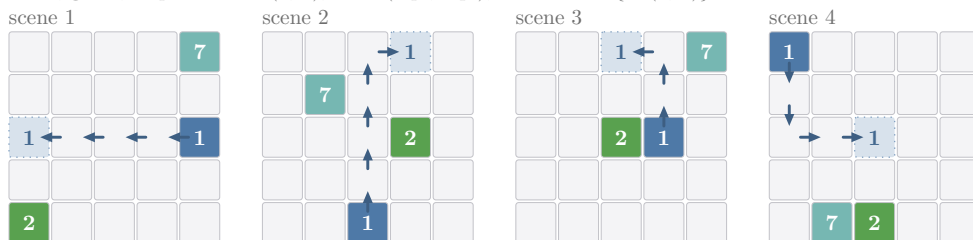
val=1



Goal displacement — Sally-Anne [belief_goal]

(fork (seq (step gv d) (optimize (neg_dist gv) av)) (sync_to_world av))

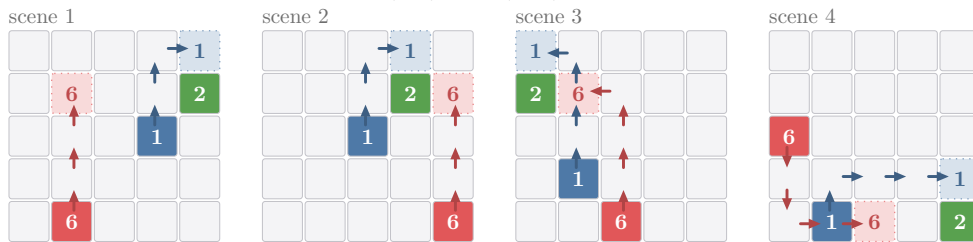
av=1, gv=2, displaced_to=(2, 0), dirs=('up', 'up'), distractors={7: (0, 4)}



Two observers — one goal, one grid, one fork [belief_observers]

(compose (optimize (neg_dist gv) ao) (fork (compose (step gv d) (optimize (neg_dist gv) av)) (sync_to_world av)))

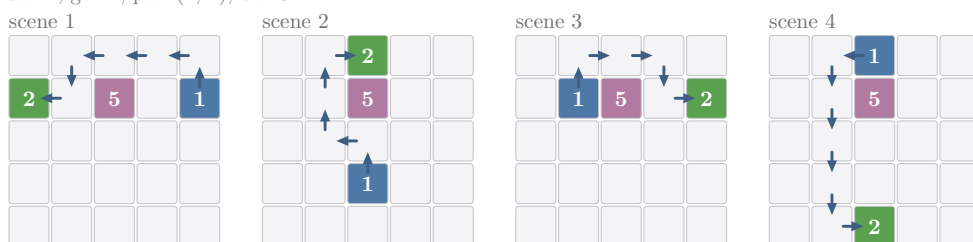
av=1, gv=2, displaced_to=(0, 4), dirs=('up',)



False belief — detour round an invisible wall [belief_wall]

(fork (compose (wall_at r c) (optimize (neg_dist gv) av)) (sync_to_world av))

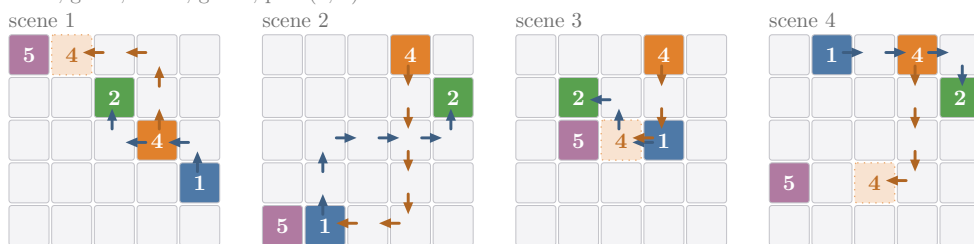
av=1, gv=2, pw=(1, 2), bv=5



Witness belief — a bystander crosses the phantom wall ^[belief_witness]

(compose (fork (compose (wall_at r c) (optimize (neg_dist gv) av)) (sync_to_world av)) (optimize (neg_dist gw) aw))

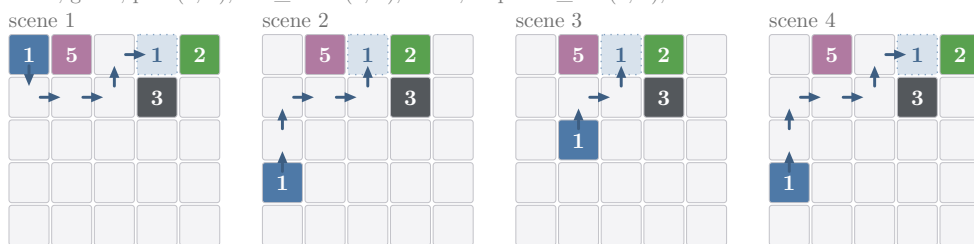
av=1, gv=2, aw=4, gw=5, pw=(1, 3)



False obstacle — wrong about wall AND goal ^[belief_false_obstacle]

(fork (seq (wall_at Wb) (step gv d) (optimize (neg_dist gv) av)) (sync_to_world av))

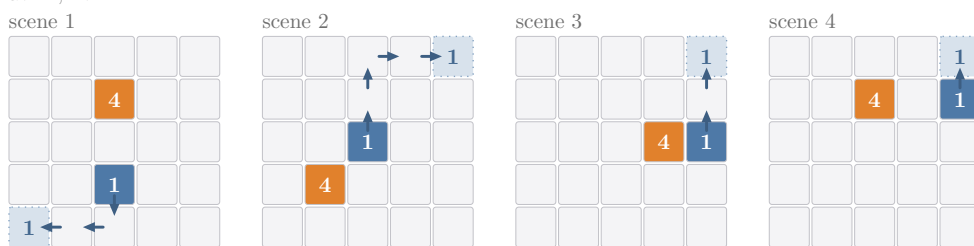
av=1, gv=2, pw=(0, 1), real_wall=(1, 3), bv=5, displaced_to=(0, 3), dir=left



Flee — avoid the nearest hazard ^[flee]

(optimize (distance hv) av)

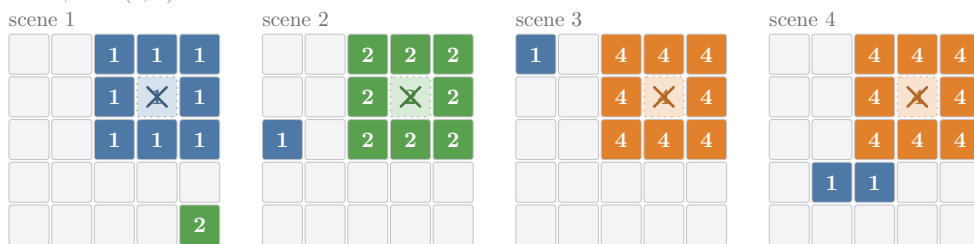
av=1, hv=4



Deletion — punch one hole ^[deletion]

(clear_at r c)

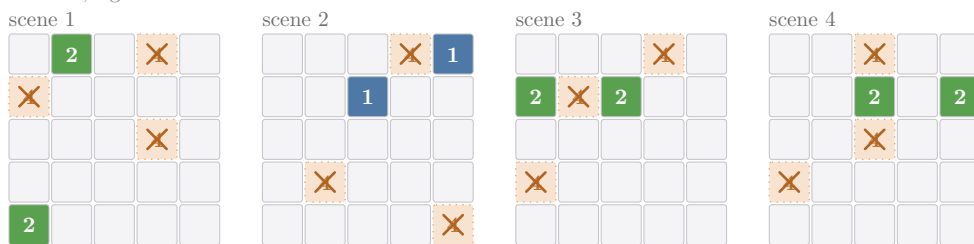
val=1, cell=(1, 3)



Denoise — drop a noise value ^[denoise]

(erase nv)

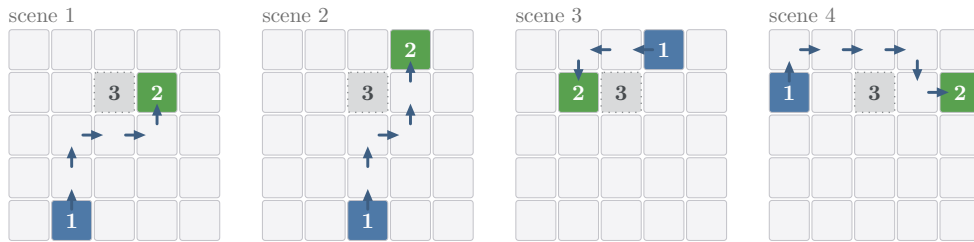
noise=4, signal=2



Obstacle — detour round a REAL wall [obstacle]

(compose (wall_at r c) (optimize (neg_dist gv) av))

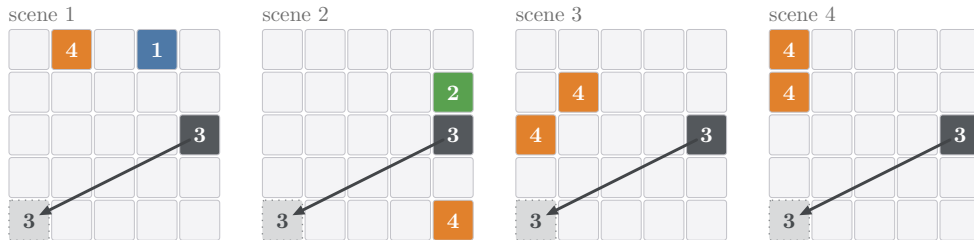
av=1, gv=2, pw=(1, 2)



Relocation — a real wall jumps cell [relocate]

(compose (clear_at r c) (wall_at r' c'))

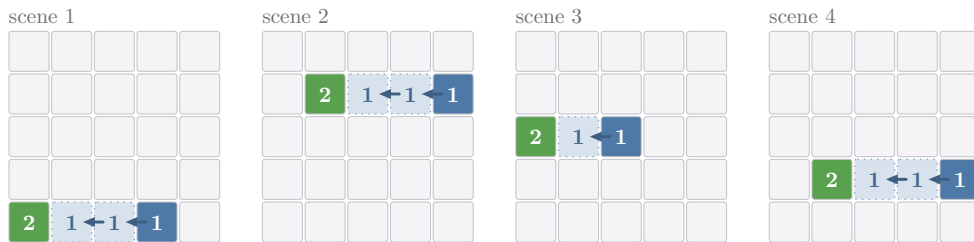
from=(2, 4), to=(4, 0)



Underlay — world-wins motion trail [underlay]

(fork (step v d) underlay)

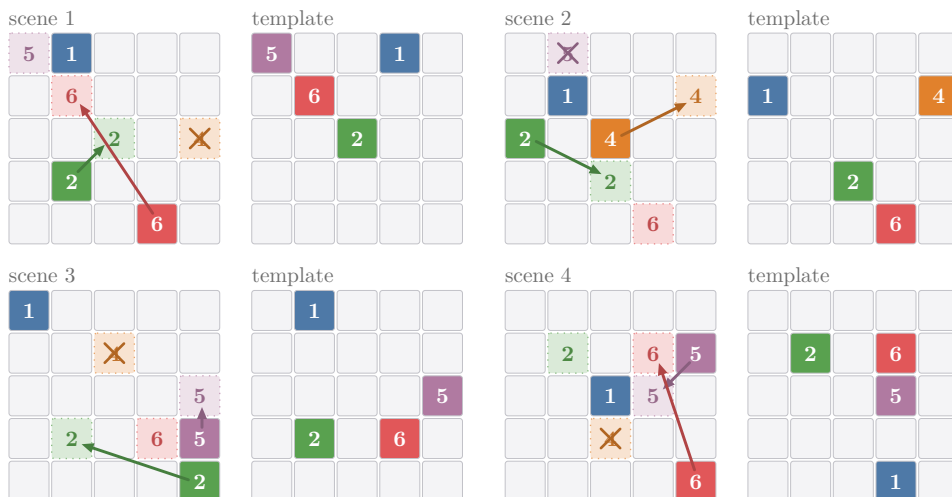
val=1, dir=left, bystander=2



Perception — record into the model [perception]

(sync_to_model v)

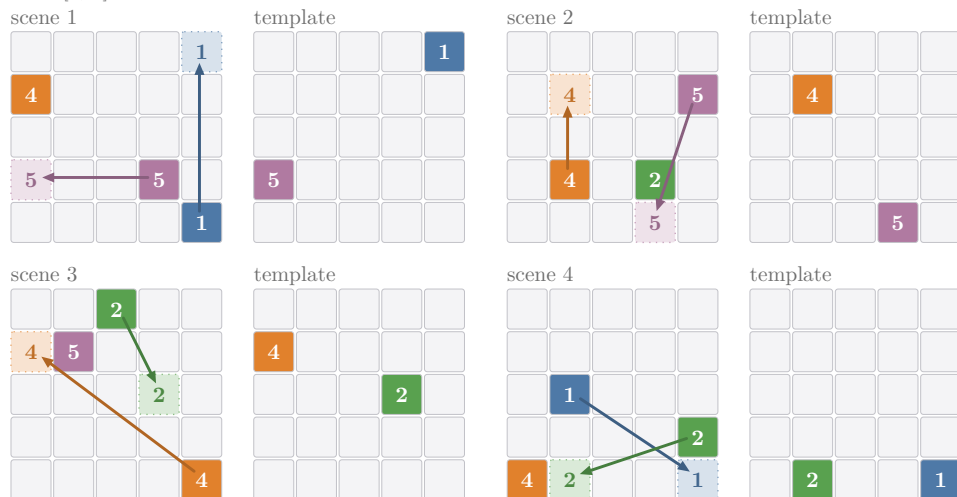
val=1



Multi-registration — snap all [multi_reg]

`sync_all`

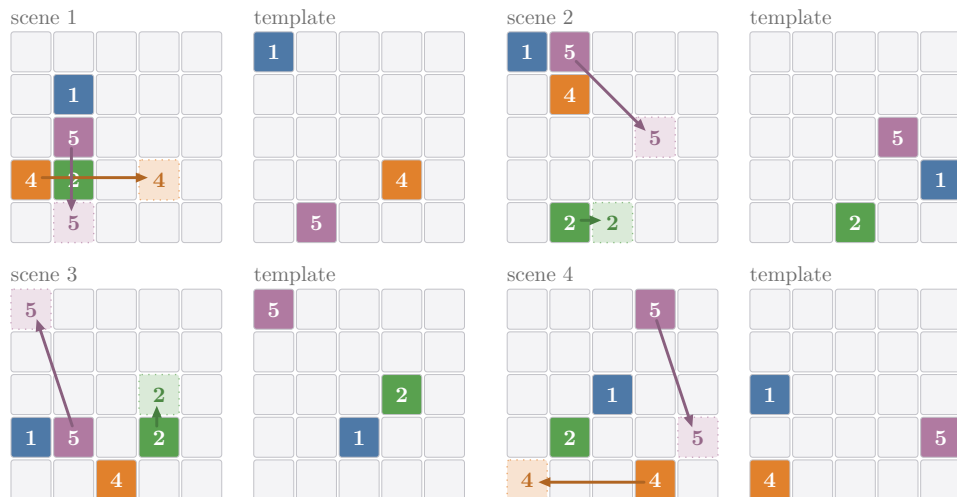
`vals=[1, 5]`



Registration-except — all but one [reg_except]

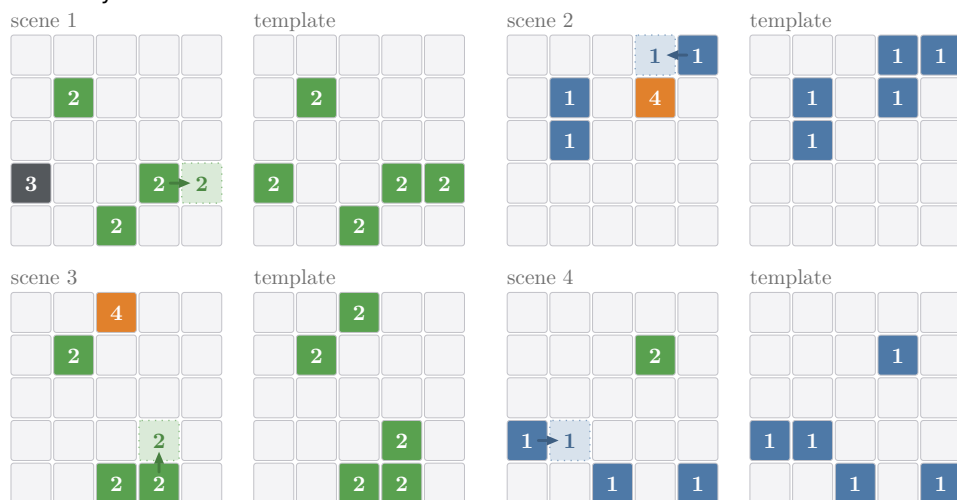
`(sync_except a)`

`anchor=1`



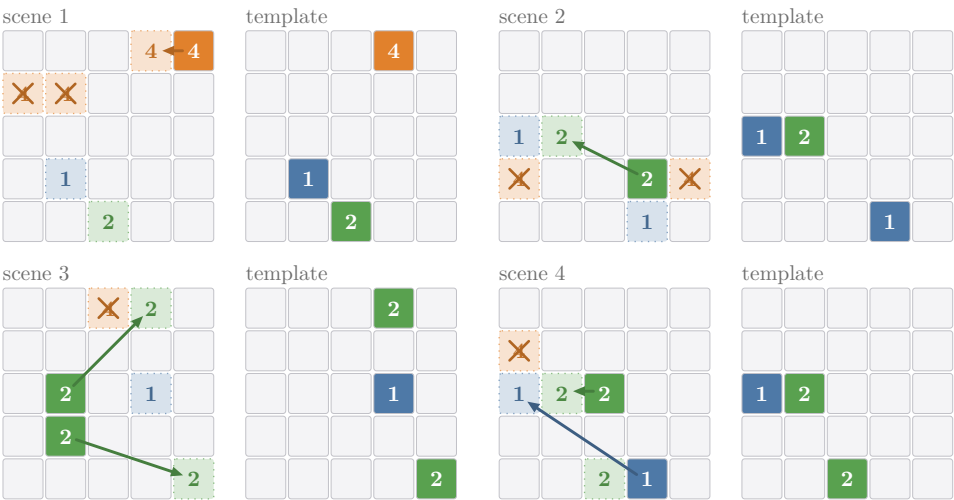
Inpainting — fill holes, keep pixels [inpaint]

`underlay`



Readout — report the model [readout]

snd_gg



Wipe — the grid goes blank [wipe]

(pipe_gpg pair_blank snd_gg)

vals=[1, 4, 6]

